Surname ______________________________

Forename(s) ______________________________

Centre Number ______________________________

Candidate Number ______________________________

Candidate Signature ______________________________

I declare this is my own work.

GCSE MATHEMATICS

F

Foundation Tier Paper 1 Non-Calculator

8300/1F

Thursday 15 May 2025 Morning

Time allowed: 1 hour 30 minutes

[Turn over]

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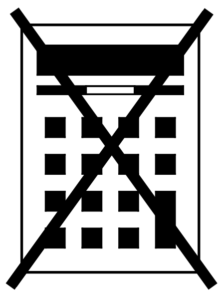
On the front of this book, write your surname and forename(s), your centre number, your candidate number and add your signature.

MATERIALS

For this paper you must have:



- **mathematical instruments**
- **the Formulae Sheet (enclosed).**

You must NOT use a calculator.

[Turn over]

INSTRUCTIONS

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer ALL questions.
- You must answer the questions in the spaces provided. Do not write on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

INFORMATION

- **The marks for questions are shown in brackets.**
- **The maximum mark for this paper is 80.**
- **You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.**

ADVICE

In all calculations, show clearly how you work out your answer.

DO NOT TURN OVER UNTIL TOLD TO DO SO

Answer ALL questions in the spaces provided.

1(a) **Write down the next number in the sequence**

1 4 7 10

[1 mark]

Answer ______________________________

1(b) Write down the next number in the sequence

2 4 8 16

[1 mark]

Answer ______________________________

[Turn over]

1(c) **Write down the next number in the sequence**

20 14 8 2

[1 mark]

Answer ______________________________

1(d) **Work out $3 \times (-6)$**
[1 mark]

Answer ______________________________

[Turn over]

4

2 Here is a card from a game.

	13		32
8		27	
	15		36

2 (a) Write down the number FROM THE CARD that is a multiple of 5
[1 mark]

Answer ______________________

2 (b) Write down the number FROM THE CARD that is a factor of 40
[1 mark]

Answer ______________________

2(c) **Write down the number FROM THE CARD that is a prime number. [1 mark]**

Answer ______________________________

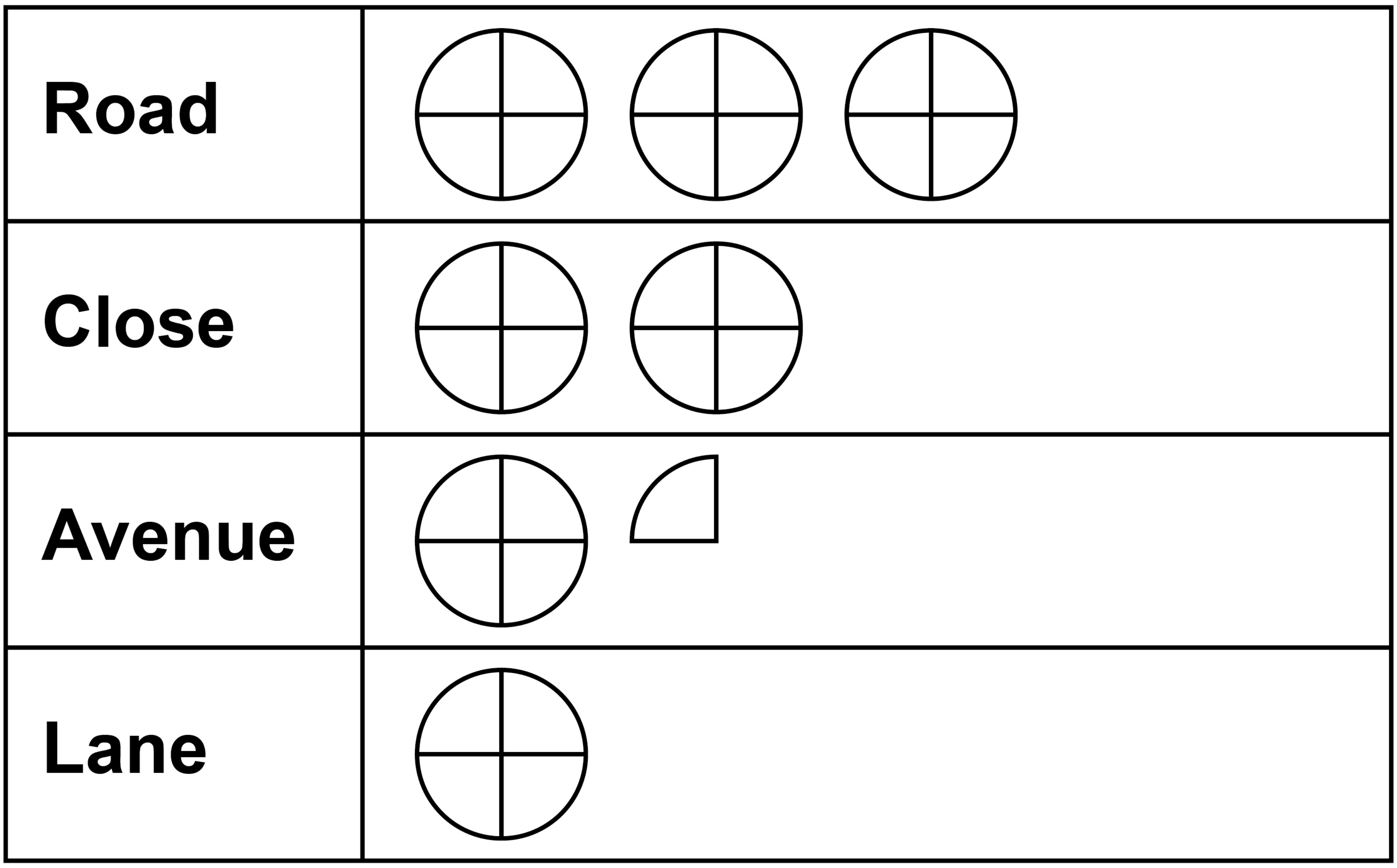
2(d) **Write down the number FROM THE CARD that is a square number. [1 mark]**

Answer ______________________________

[Turn over]

3 The pictogram shows information about the endings of street names in a town.

The key is missing.



In the town, there are 48 street names ending in CLOSE.

How many street names end in AVENUE? [3 marks]

Answer

[Turn over]

7

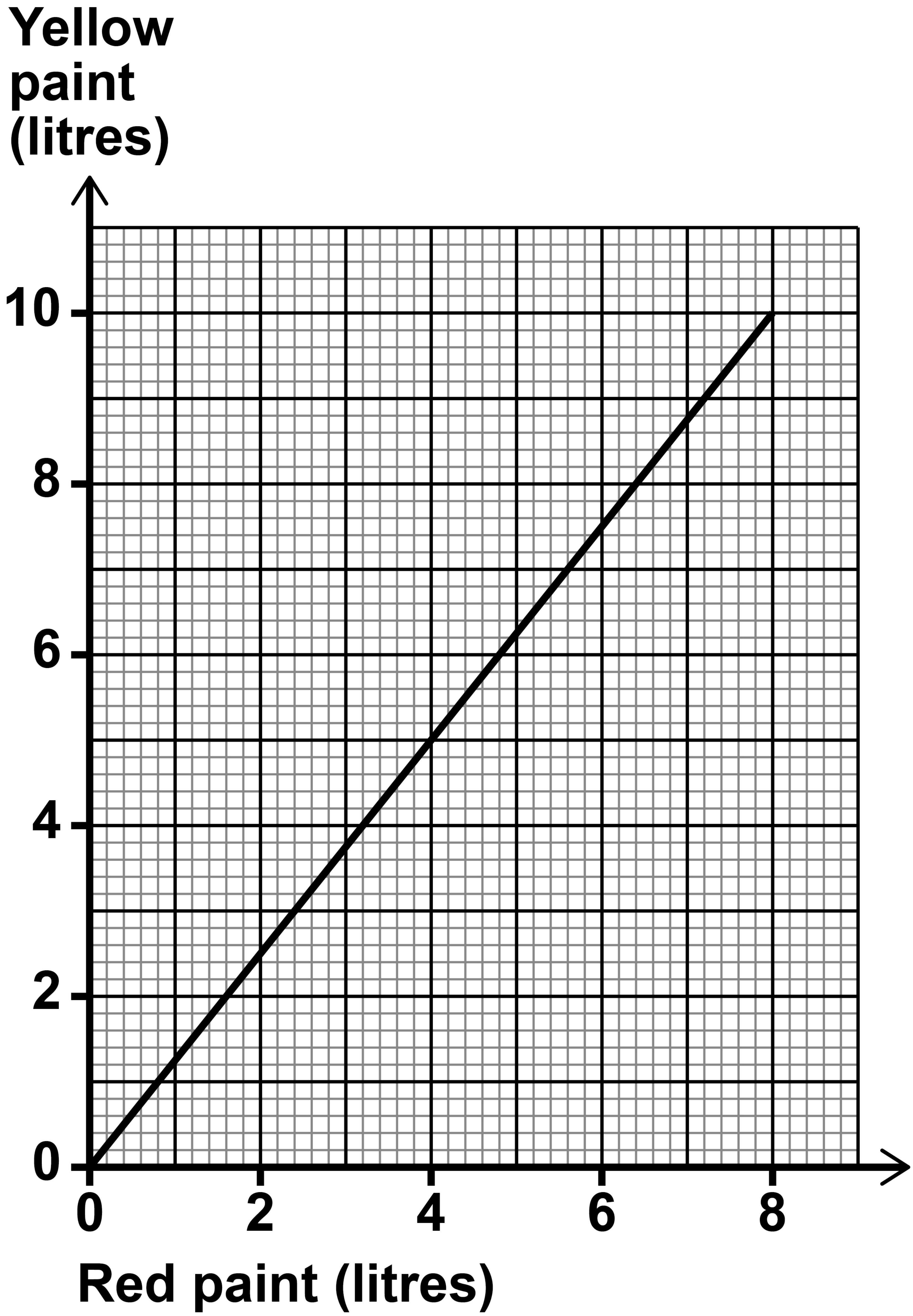
4 **Ola and Pip make orange paint by mixing red paint and yellow paint.**

The graph, on the opposite page, shows how much of each colour paint to use.

4 (a) **Ola uses 5 litres of yellow paint.**

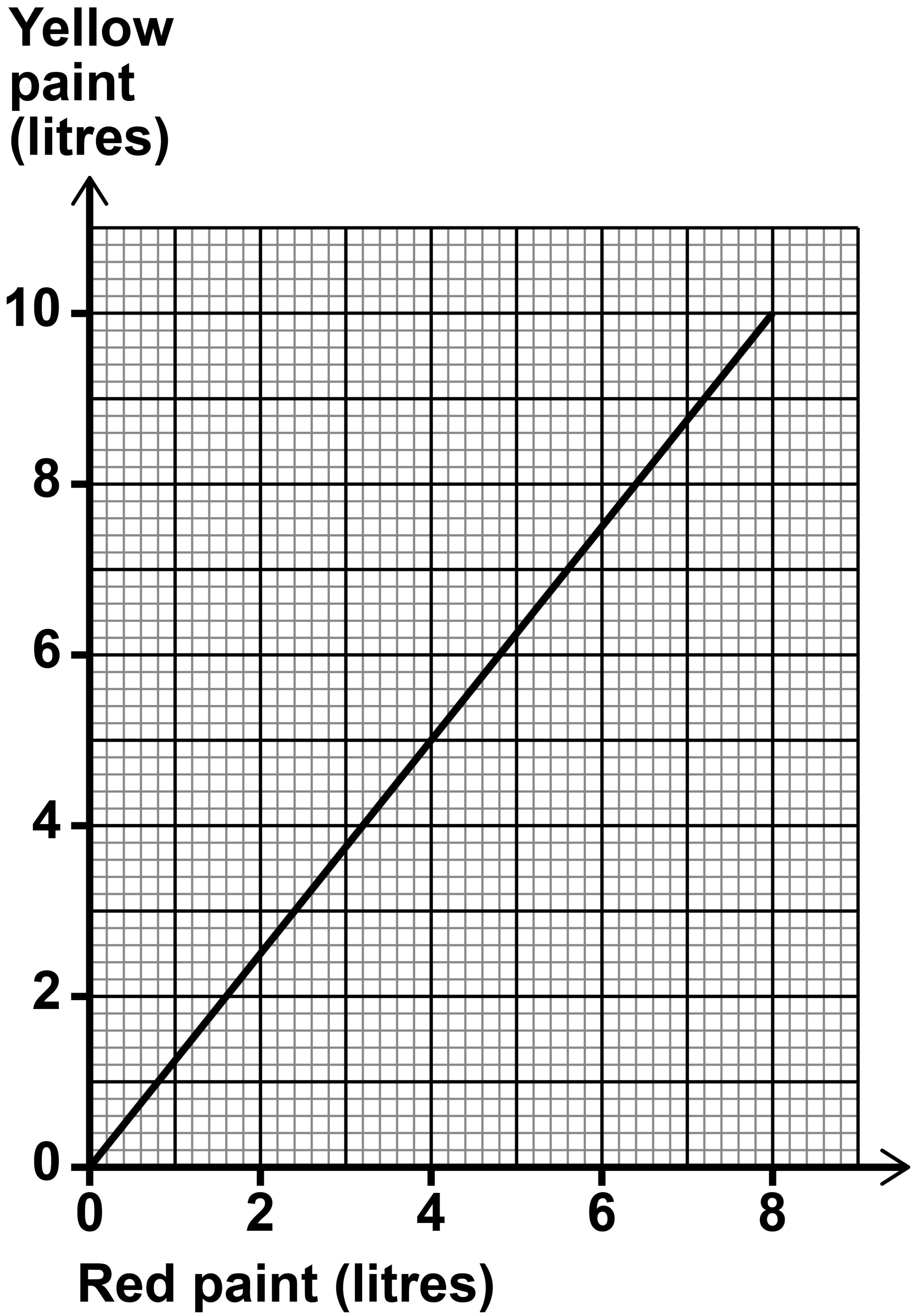
Write down how much RED paint Ola USES. [1 mark]

Answer ______________________ litres



[Turn over]

REPEAT OF GRAPH



4 (b) Pip uses 10 litres of yellow paint.

How much ORANGE paint does Pip MAKE? [2 marks]

____

____

____

Answer ____________________ litres

[Turn over]

5 **A number is divided by 8**

The answer is 43 remainder 5

Work out the number. **[3 marks]**

Answer ____________________

6 Quin buys 200 toys for £4 each.

He sells the toys for £5 each.

What is the LEAST number of toys he must sell to make a profit? [3 marks]

Answer ____________________

[Turn over]

7 Vinyl records cost £25 each.

OFFER

Buy one and get another for half price

Arif wants to buy TWO vinyl records.

He saves £8 every Saturday.

7 (a) **Assume the offer is permanent.**

How many Saturdays does Arif need to save for?

You MUST show your working.
[3 marks]

____

____

____

____

____

____

____

____

Answer ______________________

[Turn over]

7 (b) **In fact, the offer is only for 1 week and there are no new offers.**

What does this mean about the number of Saturdays he needs to save for?

Tick ONE box. [1 mark]

- [ ] **It is less than the answer to part (a)**
- [ ] **It is the same as the answer to part (a)**
- [ ] **It is greater than the answer to part (a)**
- [ ] **It is not possible to tell**

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[Turn over]

8 **Here is a list of ingredients for lentil soup.**

Lentil soup for 4 people	
Lentils	**170 g**
Water	**600 ml**
Onion	**50 g**

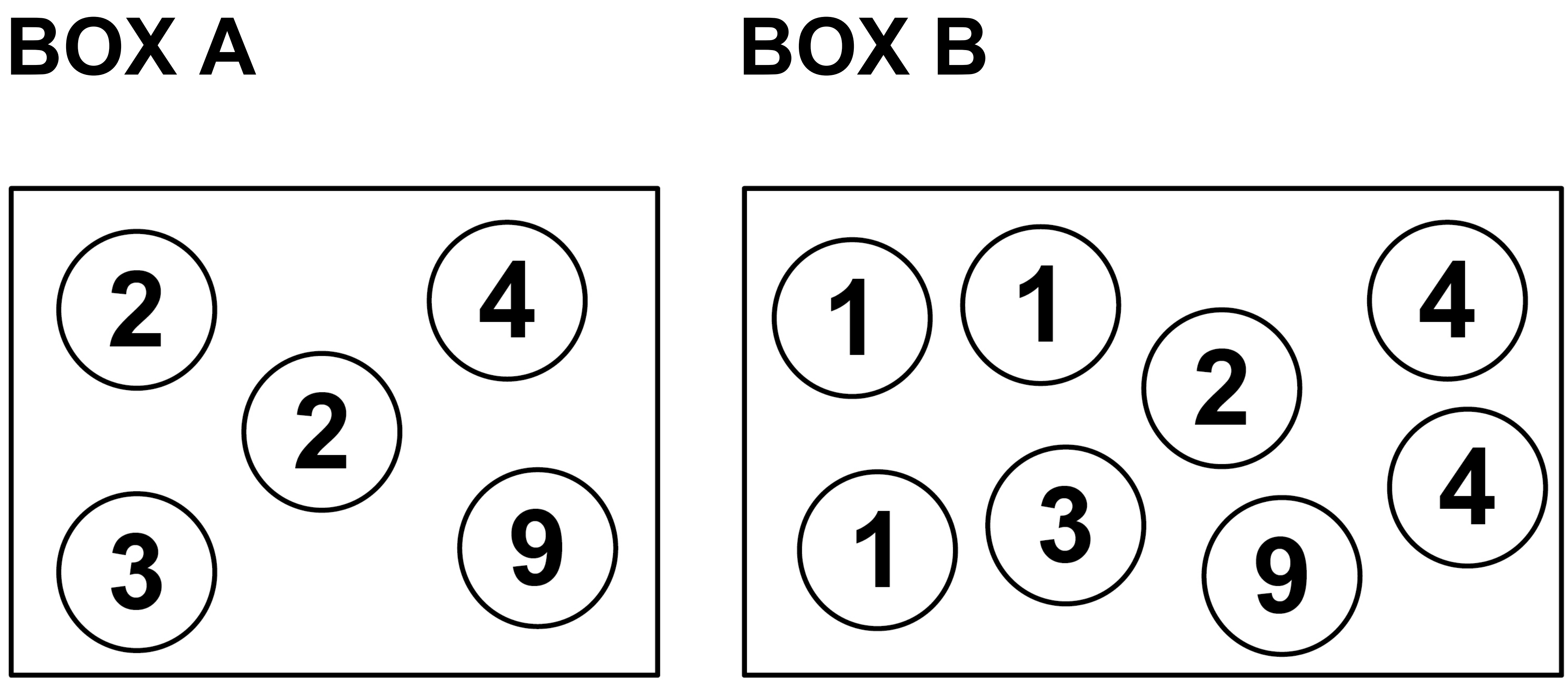
How many grams of lentils are needed to make this soup for 10 people? [3 marks]

Answer ______________________________ g

[Turn over]

7

9 Numbered discs are in two boxes.



9 (a) Work out the RATIO

total value of the numbers in BOX A :
total value of the numbers in BOX B

Give your answer in its simplest form. [3 marks]

Answer _______ : _______

[Turn over]

9 (b) **One disc is picked at random from BOX A.**

Write down the probability that the number on the disc is greater than 6
[1 mark]

Answer __________________________

10 **Work out the value of** $2(a^2 + 3a)$
when $a = 4$
[3 marks]

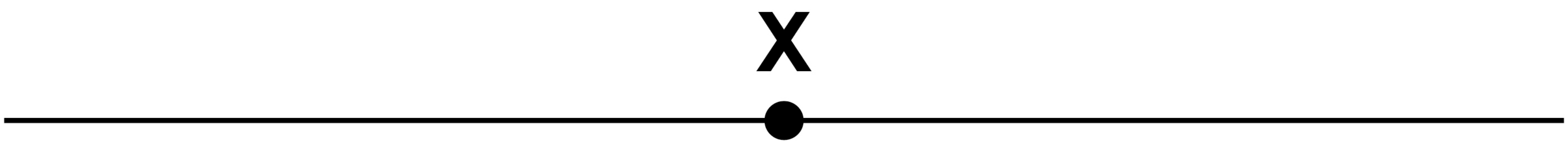
Answer ______________________

[Turn over]

11 In this question use a pair of compasses.

The line shown is the DIAMETER of a circle, centre X.

Draw the circle, on the opposite page. [1 mark]


X

12 (a) One day,

- a company runs 240 trains
- $\frac{1}{8}$ of these trains are late.

The company is charged £350 for each late train.

How much is the company charged that day? [3 marks]

Answer £ ______________________

12 (b) **Hot drinks are sold at a station in the ratio**

tea : coffee : hot chocolate = 3 : 5 : 2

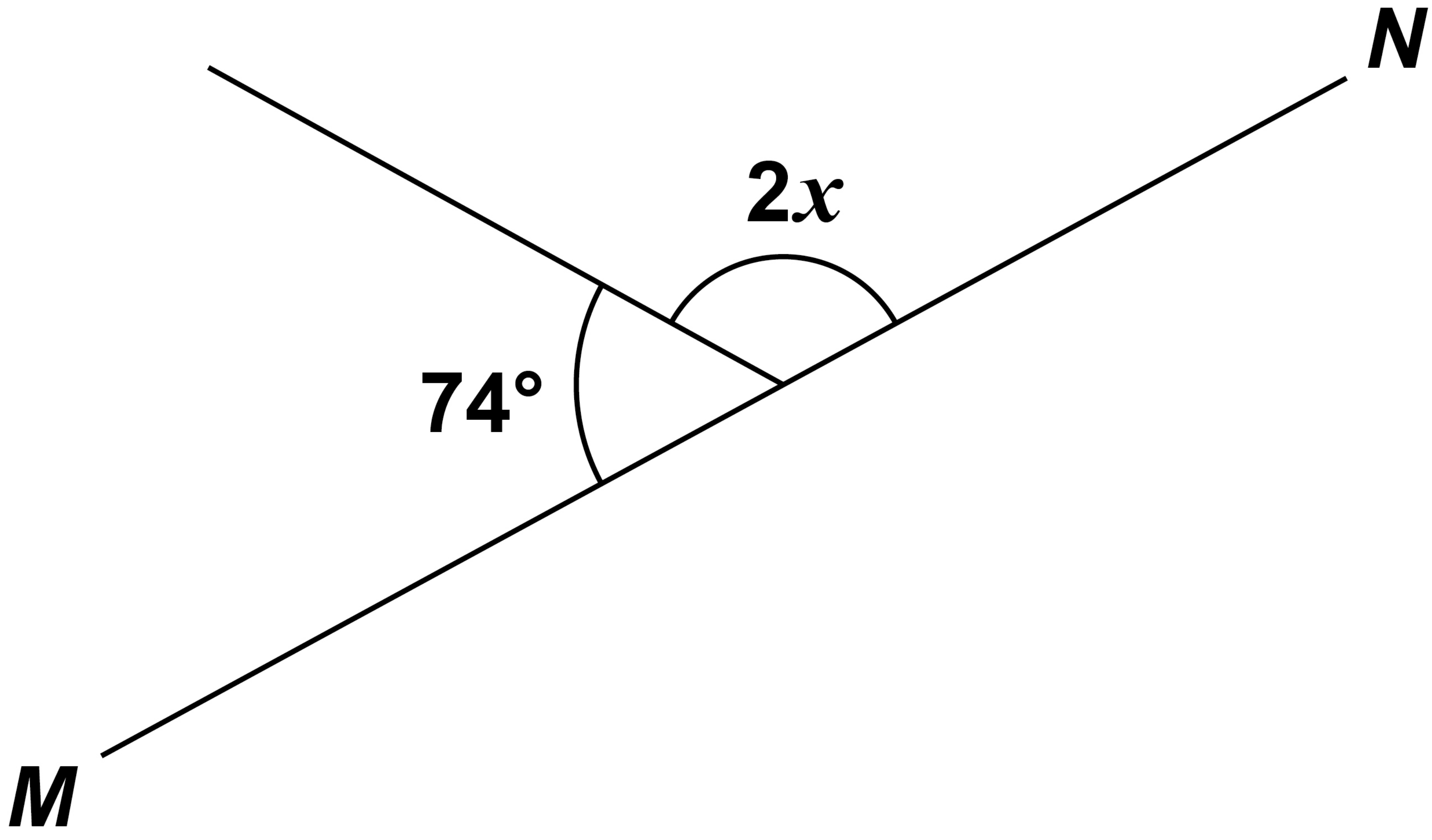
2600 hot drinks are sold.

How many coffees are sold?
[3 marks]

Answer ______________________

13 *MN* is a straight line.

The diagram is not drawn accurately.



Work out the value of x. [3 marks]

Answer ______________________ °

[Turn over]

9

14 (a) **By rounding each number to 1 significant figure, estimate the value of**

$$1.98 \times 3.82 + 6.75^2$$

You MUST show your working.
[3 marks]

Answer ____________________

14 (b) **Is your answer to part (a) an overestimate or an underestimate?**

Tick ONE box.

☐ **Overestimate**

☐ **Underestimate**

Give a reason for your answer.
[1 mark]

____

____

____

____

[Turn over]

15 (a) **A sphere has diameter 20 cm**

Show that the radius of the sphere is 10 cm
[1 mark]

15 (b) **The volume of a sphere is $\frac{4}{3}\pi r^3$**

where r is the radius.

Work out the volume of a sphere with DIAMETER 20 cm

Give your answer in terms of π
[2 marks]

____

____

____

____

Answer ______________________ **cm^3**

7

16 (a) $d = \frac{800}{w}$

- d represents the number of days to complete a job.
- w represents the number of workers needed.

Assume the job needs completing in 20 days.

How many workers are needed?
[3 marks]

Answer

16 (b) In fact, the job needs completing in FEWER than 20 days.

What does this mean about the number of workers that are needed?

Tick ONE box. [1 mark]

- [ ] It is less than the answer to part (a)
- [ ] It is the same as the answer to part (a)
- [ ] It is greater than the answer to part (a)

[Turn over]

17 **A chord is drawn on a circle.**

Which statement is correct?

Tick ONE box. [1 mark]

- [ ] **The chord must be longer than the radius.**
- [ ] **The chord must be equal in length to the radius.**
- [ ] **The chord must be shorter than the radius.**
- [ ] **The chord could be longer than, equal in length to or shorter than the radius.**

18 **A metal solid has volume 11 cm^3**

The density of the metal is 8.5 g/cm^3

Work out the mass of the solid.
[2 marks]

Answer ______________________________ **g**

[Turn over]

7

19 The table shows information about the marks of students in a test.

	MEAN MARK	RANGE OF MARKS
SCHOOL A	61	14
SCHOOL B	56	21

Tick ONE box for each statement, on the opposite page. [3 marks]

	TRUE	MAY BE TRUE	NOT TRUE
On average, School A had higher marks	☐	☐	☐
There are more students in School B	☐	☐	☐
School B had a greater spread of marks	☐	☐	☐

[Turn over]

20 (a) **Work out $0.6 \div 100$**

Give your answer in standard form. [2 marks]

Answer ______________________

20 (b) Work out $40 \times 30 \times 10^5$

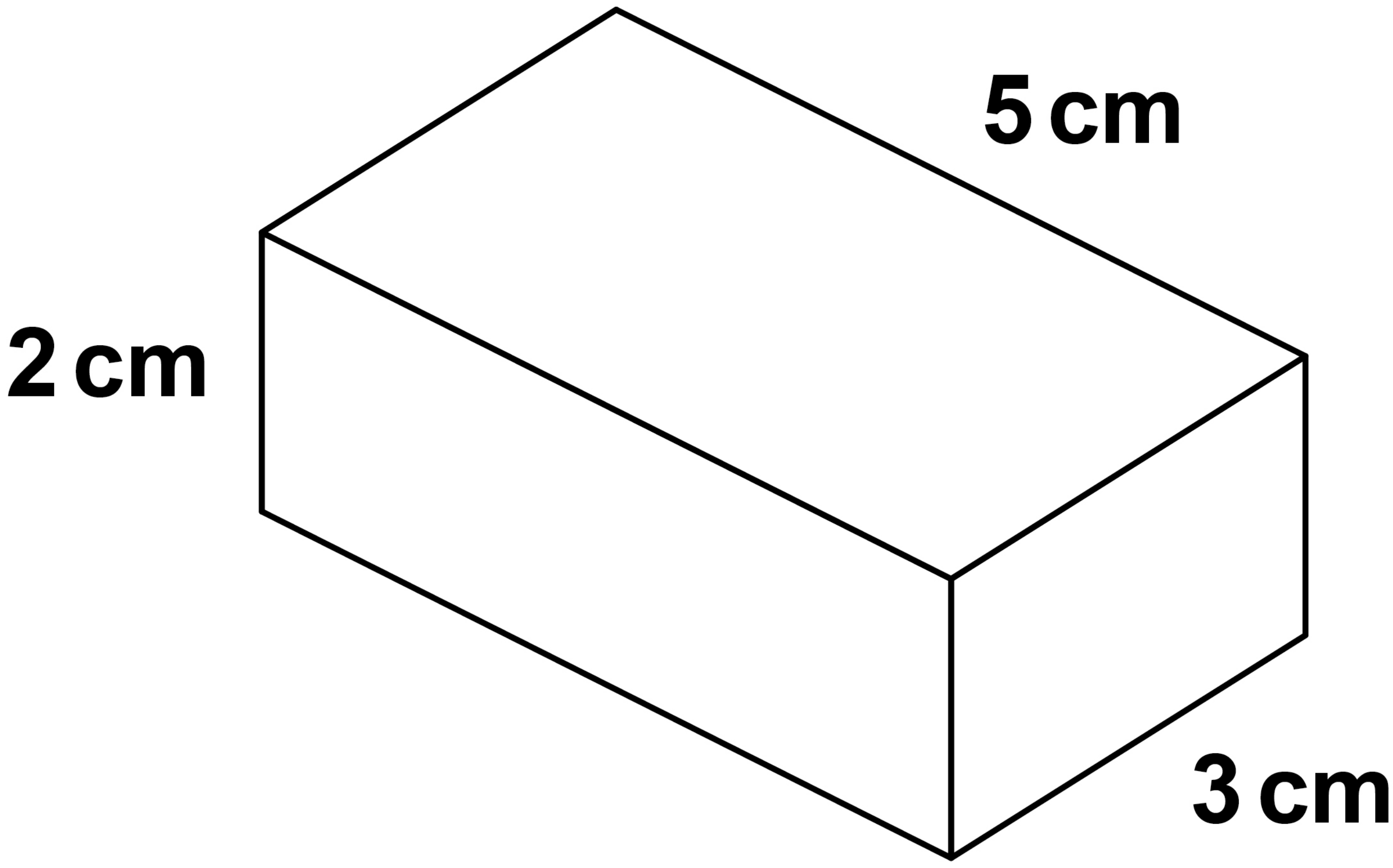
Give your answer in standard form. [2 marks]

Answer ______________________

[Turn over]

7

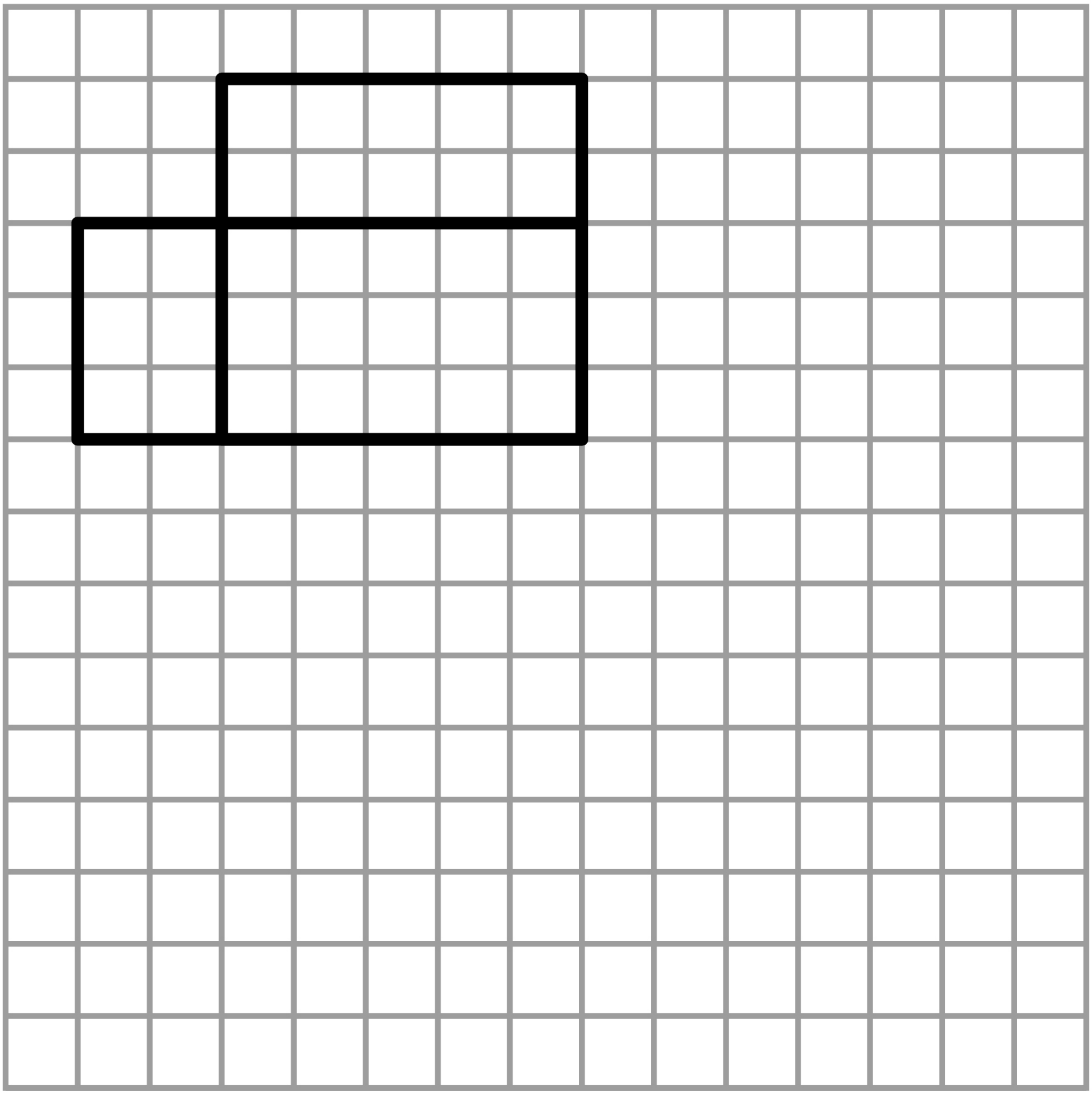
21 Here is a cuboid.



On the opposite page, complete the drawing of the net on the grid.
[2 marks]

The grid is NOT drawn to scale.

Use a scale of 1 square = 1 cm

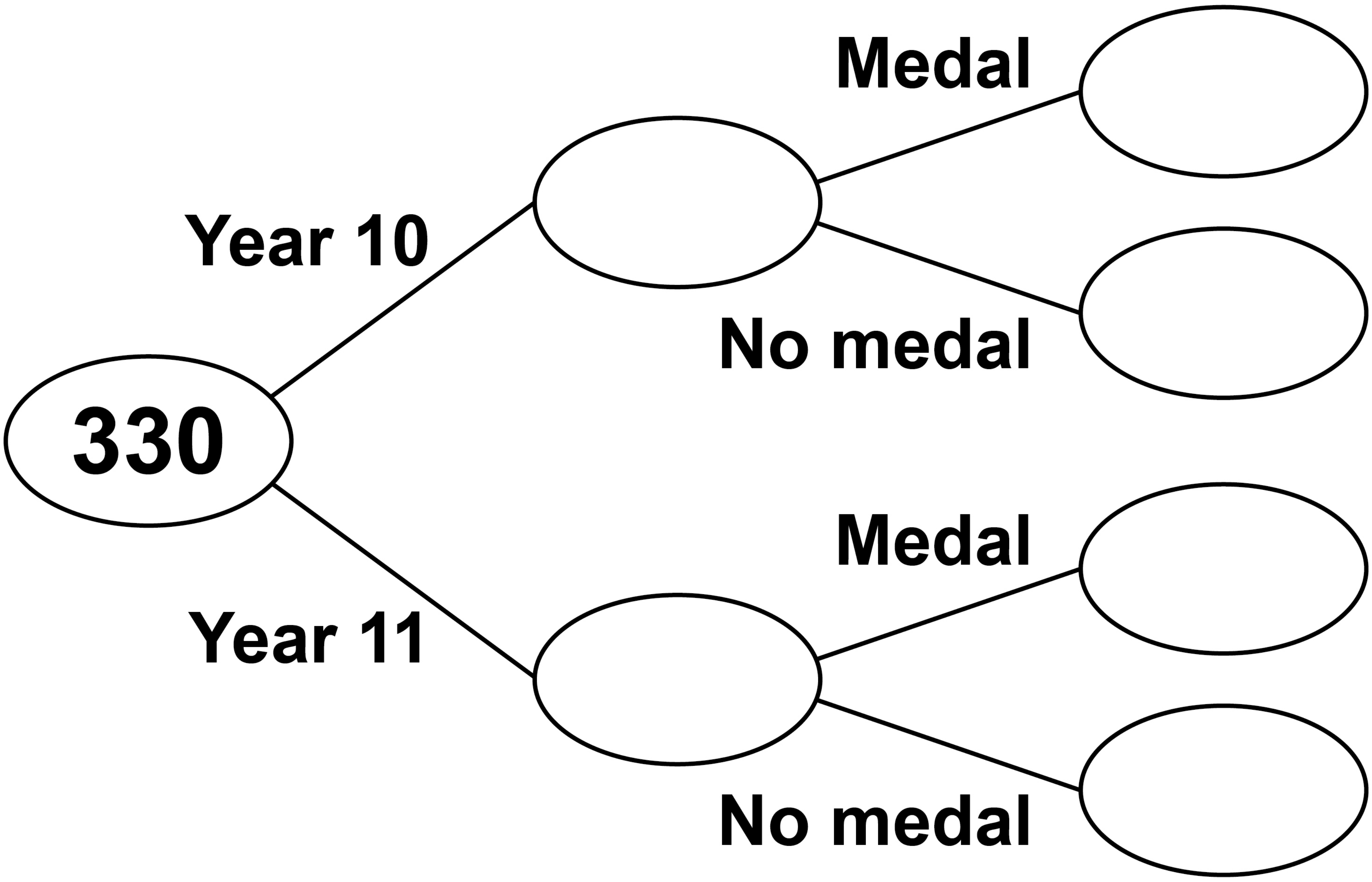


[Turn over]

22 330 students from Year 10 and Year 11 take part in a competition.

- **number of students in Year 10 : number of students in Year 11 = 1 : 2**
- **125 students win a medal.**
- **73 of the students who win a medal are in Year 11**

Complete the frequency tree, on the opposite page. [4 marks]



6

23 **Work out** $\frac{4}{15} + \frac{1}{5} \div \frac{1}{2}$

Give your answer as a fraction.
[3 marks]

Answer ______________________

24 $y = \dfrac{1}{x}$

Which of these values of x gives the GREATEST value of y ?

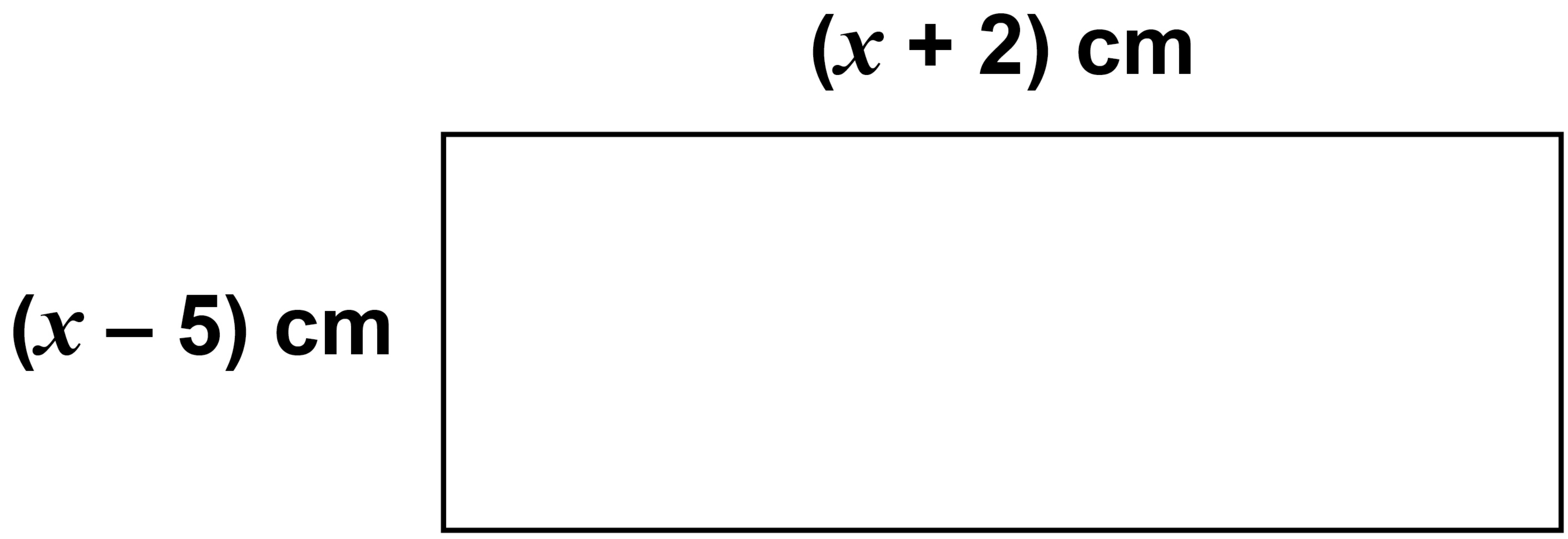
Circle your answer. [1 mark]

$\dfrac{9}{20}$ $\dfrac{2}{5}$ -80 95

25 Circle the value of $\sin 90°$
[1 mark]

0 $\dfrac{1}{2}$ $\dfrac{\sqrt{3}}{2}$ 1

26 **The diagram is not drawn accurately.**



The area of the rectangle is 120 cm^2

Work out the value of x. [4 marks]

x = ______________________

END OF QUESTIONS

$\frac{}{9}$

Additional page, if required.

Write the question numbers in the left-hand margin.

Additional page, if required.

Write the question numbers in the left-hand margin.

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For Examiner's Use	
Pages	Mark
6–9	
10–13	
14–19	
20–25	
26–31	
32–35	
36–39	
40–43	
44–47	
48–51	
52–55	
TOTAL	

WP/M/SB/Jun25/8300/1F/G4010/V5